

Yijun Liu

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🏠 [Personal Homepage](#)

🎓 👤 Education

Southern University of Science and Technology (SUSTech) · Computer Science and Technology (Undergraduate) August 2024 – June 2028

GPA 3.73/4.0, received university scholarship. Scored 670 in National College Entrance Exam (Gaokao), ranking 715th among 203,000 candidates in Chongqing. Core courses: Artificial Intelligence, Algorithm Analysis and Design, Data Structures and Algorithm Analysis, Engineering Probability and Statistics, Calculus, Linear Algebra.

🎓 ⚙️ IT Skills

- Programming Languages: Python (proficient), Java (proficient), SQL (familiar);
- Development Tools: Git, VS Code, Mac/Linux/Windows environments;
- Finance Fundamentals: basic knowledge of stocks, funds, bonds;
- Others: English (proficient, TOEFL 100).

🎓 🧪 Research Direction

Factor Mining · Under Professor Peng Yang

September 2025 – Present

Focused on the intersection of machine learning and quantitative investment. Main directions include:

- **Reinforcement Learning (RL) Factor Mining:** Using RL to automatically mine alpha factors with high predictive power, building end-to-end stock trend prediction models;
- **Integration of RL with Other Methods:** Combining RL with optimization methods such as genetic algorithms to design efficient training frameworks, improving factor mining speed and generalization;
- **Cross-Domain Technology Transfer:** Adapting advanced techniques from robotics, computer vision, etc. (e.g., diffusion models, representation learning, imitation learning) to improve discrete alpha factor generation.

🎓 📁 Research Project Experience

Evolutionary Strategy and Information Ratio Shaping Reward for Quantitative Factor Generation Framework January 2026 – March 2026

Research project (novel factor mining method integrating RL and genetic algorithms) · Python

- Integrates RL with genetic algorithms, directly optimizing strategies in parameter space via evolutionary strategy, avoiding backpropagation and greatly improving efficiency;
- Mitigates gradient variance explosion under hundreds of thousands of dimensions using mirror sampling and gradient clipping;
- Designs a dedicated reward function to replace short-term IC, effectively suppressing noisy factors overfitted to short-term IC and enhancing factor stability on test sets.

Research project (novel factor mining paradigm based on diffusion models) · Python

- Extends diffusion models to discrete alpha factor mining, achieving end-to-end differentiable non-autoregressive factor generation;
- Solves continuous embedding and decoding of discrete factors by incorporating trainable embeddings to map tokens into vectors, significantly improving decoding validity;
- Overcomes sparse distribution caused by small-sample expert factor libraries, using semantic enhancement and drift repulsion to avoid mode collapse.